Drilling Mechanics - APWD Time Based Real Time, Recorded Mode

Schlumberger

Company:	Equinor
Well:	31/5-7
Field:	Eos
Rig Name:	West Hercules
Country:	Norway

Latitude:	60° 34' 35.14" N	UWID:	
Longitude:	3° 26' 36.13" E	Rig Name:	West Hercules
Block:	31/5	Rig Type:	Semi-Submersible
FL:	Norwegian North Sea		
FL1:	Section: 12 1/4 in. pilot		
FL2:	Job no.: 19SCA0085		

Log Measured From: - Drill Floor: 31.00 m
Permanent Datum: - Mean Sea Level
Ground Level: 307.00 m

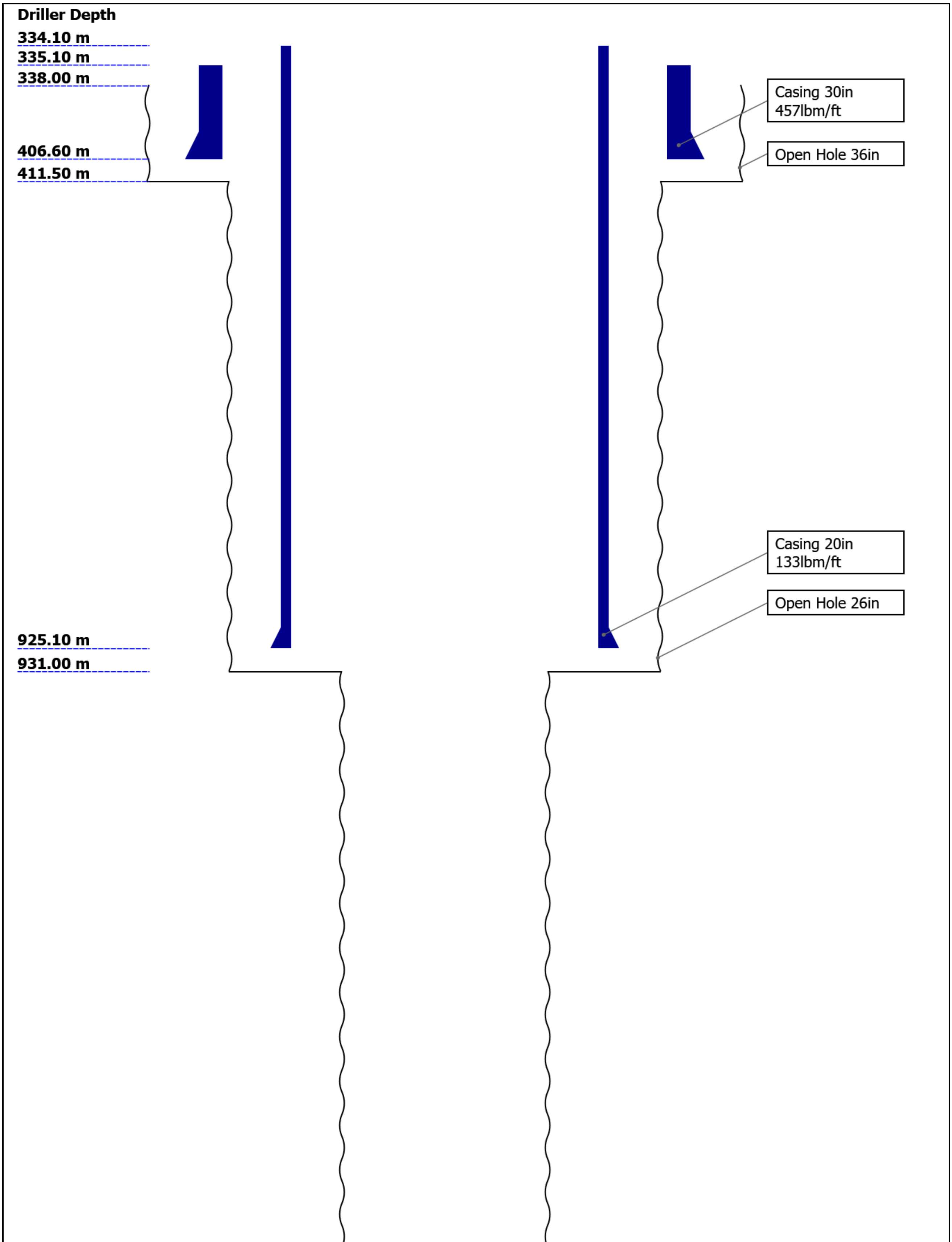
Acquisition Dates:	16-Dec-2019 -- 17-Dec-2019
Log Interval:	12/16/2019 9:45:00 AM -- 12/17/2019 5:45:00 PM
Index Types:	Time
Index Scales:	5 cm / 3600 secs
Depth Source:	Driller's Depth
Depth Sensor:	3rd Party Depth
Print Type:	Final
Spud Date:	02-Dec-2019

Other Services:
Direction & Inclination (Surveys)
Directional Drilling
VISION Resistivity

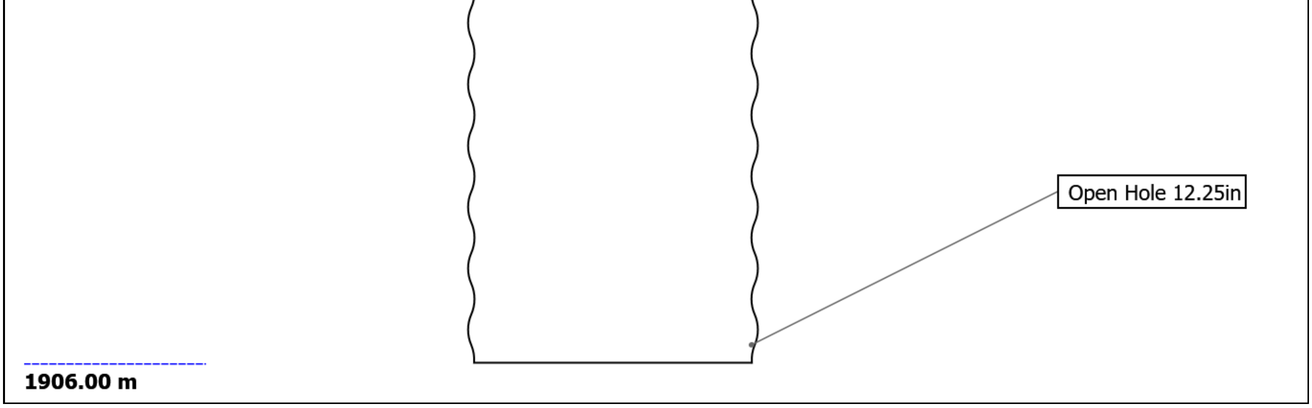
Disclaimer

Contents

Well Sketch



Driller Depth
334.10 m
335.10 m
338.00 m
406.60 m
411.50 m
925.10 m
931.00 m
Casing 30in 457lbm/ft
Open Hole 36in
Casing 20in 133lbm/ft
Open Hole 26in



Borehole Size/Casing Record

Bit							
Bit Size (in)	36	26	12.25				
Top Driller (m)	338	411.5	931				
Bottom Driller (m)	411.5	931	1906				
Casing							
Size (in)	30	20					
Weight (lbm/ft)	457	133					
Inner Diameter (in)	27.067	18.73					
Grade	X56	N80					
Top Driller (m)	335.1	334.1					
Bottom Driller (m)	406.6	925.1					

Operational Run Summary

Parameter (unit)	**Run 5**						
Date Log Started	16-Dec-2019						
Time Log Started	09:06:10						
Date Log Finished	17-Dec-2019						
Time Log Finished	18:13:52						
Bit Size (in)	12.250						
Bit Start Depth (m)	931.00						
Bit Stop Depth (m)	1906.00						
Top Log Interval (m)	925.80						
Bottom Log Interval (m)	1896.80						
Max Hole Deviation (deg)	1.02						
Azimuth of Max Deviation (deg)	139.83						
Logging Unit Number	A615						
Logging Unit Location	Service Office						
Recorded By	G.Boyd						
Witnessed By	S.Vikra						
Service Order Number	19SCA0085						

Borehole Fluids

Parameter(unit)	Run 5						
Fluid Type	Water						
Fluid Name	Glydril						
Max Recorded Temperatures (degC)	34						
Source of Sample	Active Tank						
Salinity (ppm)	Zoned						
Density (g/cm3)	1.25						
Funnel Viscosity (s)							
Fluid Loss (cm3)							
PH							
Source RMF							
RMC	Pressed						
RM @ Meas Temp (ohm.m@degC)	0.08 @ 17.78						
RMF @ Meas Temp (ohm.m@degC)	0.07 @ 18.6						
RMC @ Meas Temp (ohm.m@degC)	0.09 @ 20						
RM @ BHT (ohm.m@degC)	0.05 @ 34						
RMF @ BHT (ohm.m@degC)	0.05 @ 34						
RMC @ BHT (ohm.m@degC)	0.07 @ 34						
Total Solid (%)							
High Gravity Solids (%)							

Zoned Borehole Fluids

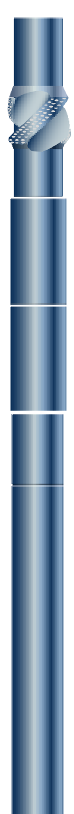
Run 5

Parameter	Value	Start
Salinity	114390.2	12/16/2019 9:06:10 AM
Salinity	114390.2	12/17/2019 11:56:31 AM

Remarks and Equipment Summary

Run 5: Toolstring

Equip name	Length	MP name	Offset
Stabilizer:MWS 37 32	**24.59**	12 1/8 in. Non-Mag Stab	
X/O: 9 1/2"[2]:XO S-947	**22.09**	X-Over	
X/O: 9 1/2"[1]:71 539	**21.26**	X-Over	
TELE900:G5093 MSSU900:71539 Upper Extender:26-0 2-16 MDC900:G5093 MMA:2068 MDI:3956 PMGR:455 PMEA:3708 MTA:2526 MTK900:008/298895 -2142 MSSD900:973247-1 Lower Extender:12/1 42	**20.62**	TeleScope	



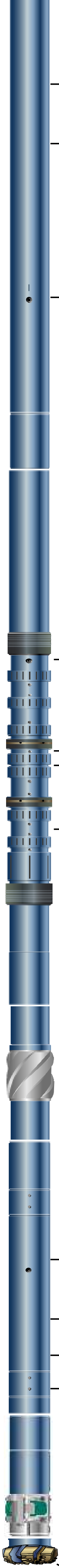
Run 5: Remarks

All depths referenced to driller's depth. See EOWR for depth tracking details.

APWD data is from tool memory. All other data is Real Time.

Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and Potassium content.

Run 5 and 12 1/4 in. pilot section TD at 1906 m.



D&I 16.69

GR 16.05

ROP 14.34

ARC9:5969 **12.58** arcVISION
ARDC:5969
Upper Extender:07-1
84
CDJA
AREA:8374
APWD:1820
ARSS:105209
Lower Extender:13/0
13

ROP 10.31

GR 9.24

Resistivity 9.19

Pressure 8.48

PDORBIT_900:98
712S **6.81** PowerDrive Orbit
PD9SRX:83196
PD9SRX Sleeve
PD9SRC:70550
PD900CC:50552
Gamma Set:37072
PDCU:606
Impeller Set
PDORB900BU:98712
S

Azimuth 3.36

D&I 2.69

GR 2.44

Anchor Bolt 1.71

Bit: 12 1/4" **0.3** Smith PDC MDSiZ616 Stringer

TOOL_ZERO

Lengths are in m

Maximum Outer Diameter = 12.250 in
Line: Sensor Location, Value: Gating Offset
All measurements are relative to TOOL_ZERO

Run 5

Drilling Mechanics - APWD - Time Based

Pass Summary

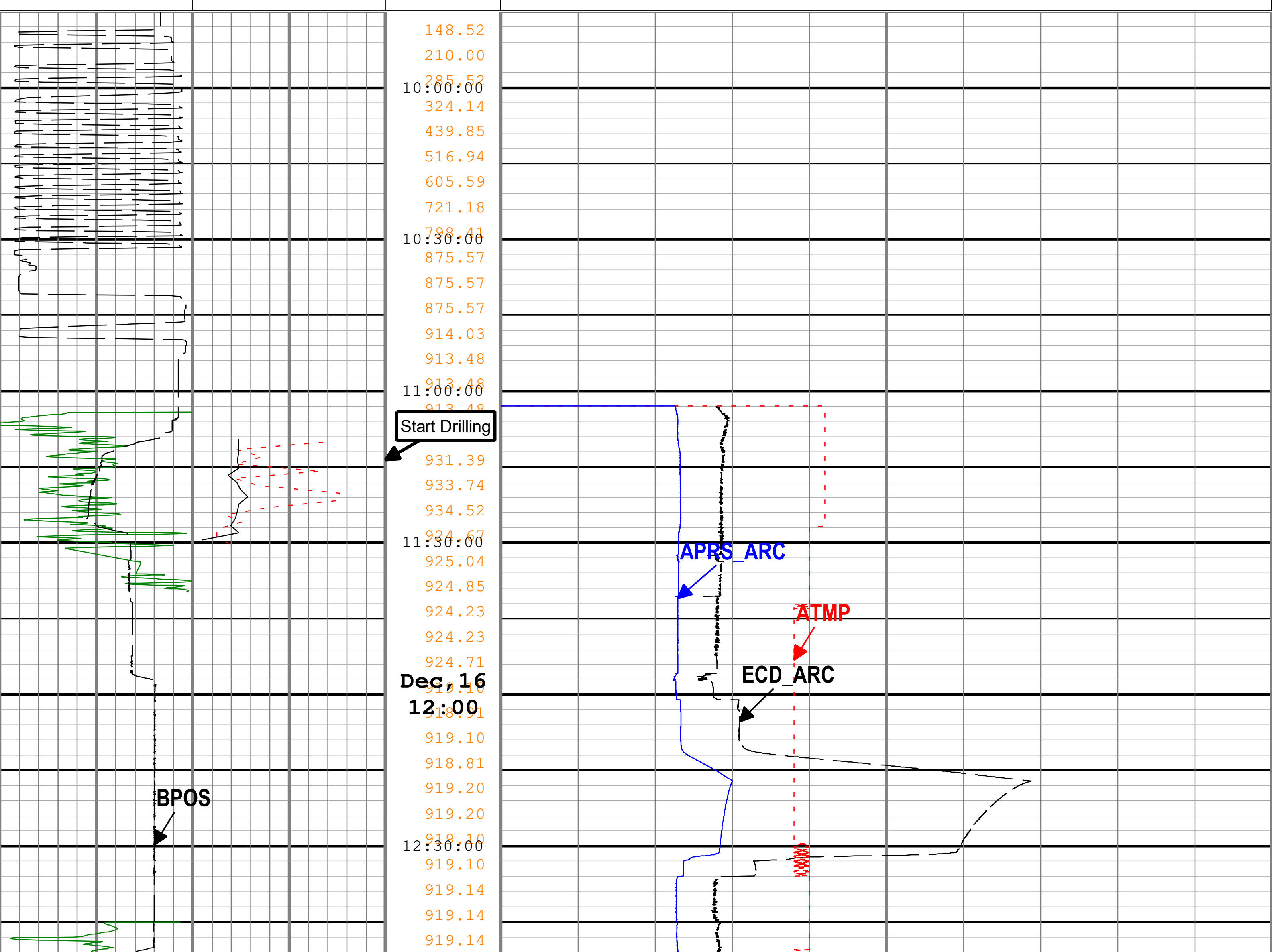
Run Name	Pass Objective	Direction	Start	Stop	Include Parallel Data
Run 5	TimeLogicalAcq	Down	16-Dec-2019 9:06:10 AM	17-Dec-2019 6:13:52 PM	Yes

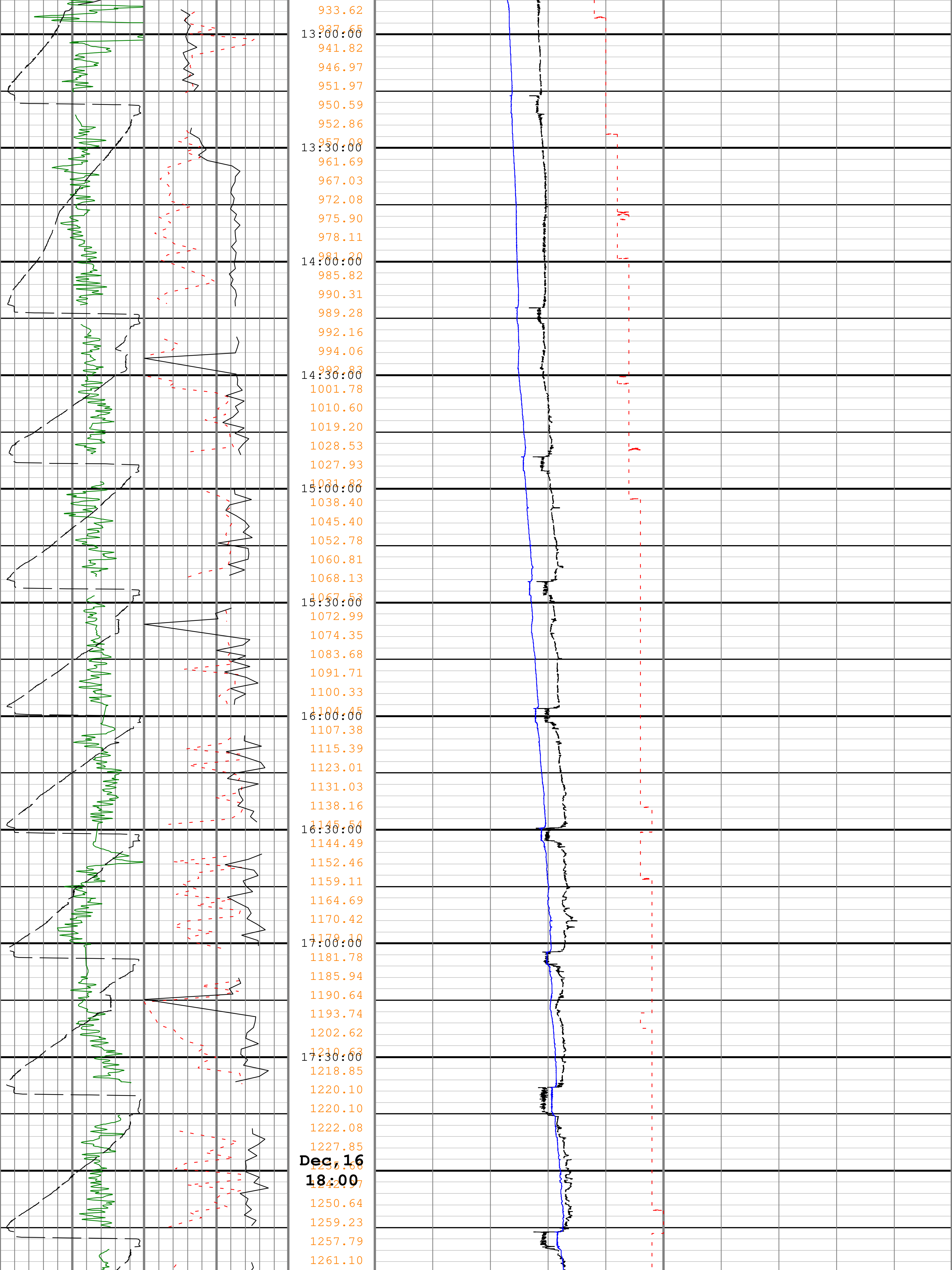
Log

Company:Equinor Well:31/5-7
Run 5: TimeLogicalAcq:S061

Description: Format: Log (Drilling Mechanics Light) Index Scale: 5 cm per 3600 s Index Type: Time Creation Date: 20-Dec-2019 14:32:04

Track 1	Track 2	Depth track	Track 3
Gamma Ray (GR_ARC_RT) ARC9 RT: 0 gAPI 150	Stick Slip Indicator (STICK_RT) TELE900 RT: 0 c/min 500	Depth Index (DEPTH) RT m	Equivalent Circulating Density (ECD_ARC) ARC9 RM: 1 g/cm3 2
Height of block above rig floor (BPOS) RT: 0 m 45	Collar Rotational Speed (CRPM_RT) TELE900 RT: 0 c/min 250	Shock Risk (SHKRSK_RT) TELE900 RT: 5 0	Downhole Annulus Temperature (ATMP) ARC9 RM: 0 degC 50
		Shock Peak (SHKPK_RT) TELE900 RT: 0 gn 250	Downhole Annulus Pressure (APRS_ARC) ARC9 RM: 0 bar 500





933.62
937.65
13:00:00
941.82
946.97
951.97
950.59
952.86
957.09
13:30:00
961.69
967.03
972.08
975.90
978.11
981.20
14:00:00
985.82
990.31
989.28
992.16
994.06
992.83
14:30:00
1001.78
1010.60
1019.20
1028.53
1027.93
1031.82
15:00:00
1038.40
1045.40
1052.78
1060.81
1068.13
1067.53
15:30:00
1072.99
1074.35
1083.68
1091.71
1100.33
1104.45
16:00:00
1107.38
1115.39
1123.01
1131.03
1138.16
1145.54
16:30:00
1144.49
1152.46
1159.11
1164.69
1170.42
1179.10
17:00:00
1181.78
1185.94
1190.64
1193.74
1202.62
1210.63
17:30:00
1218.85
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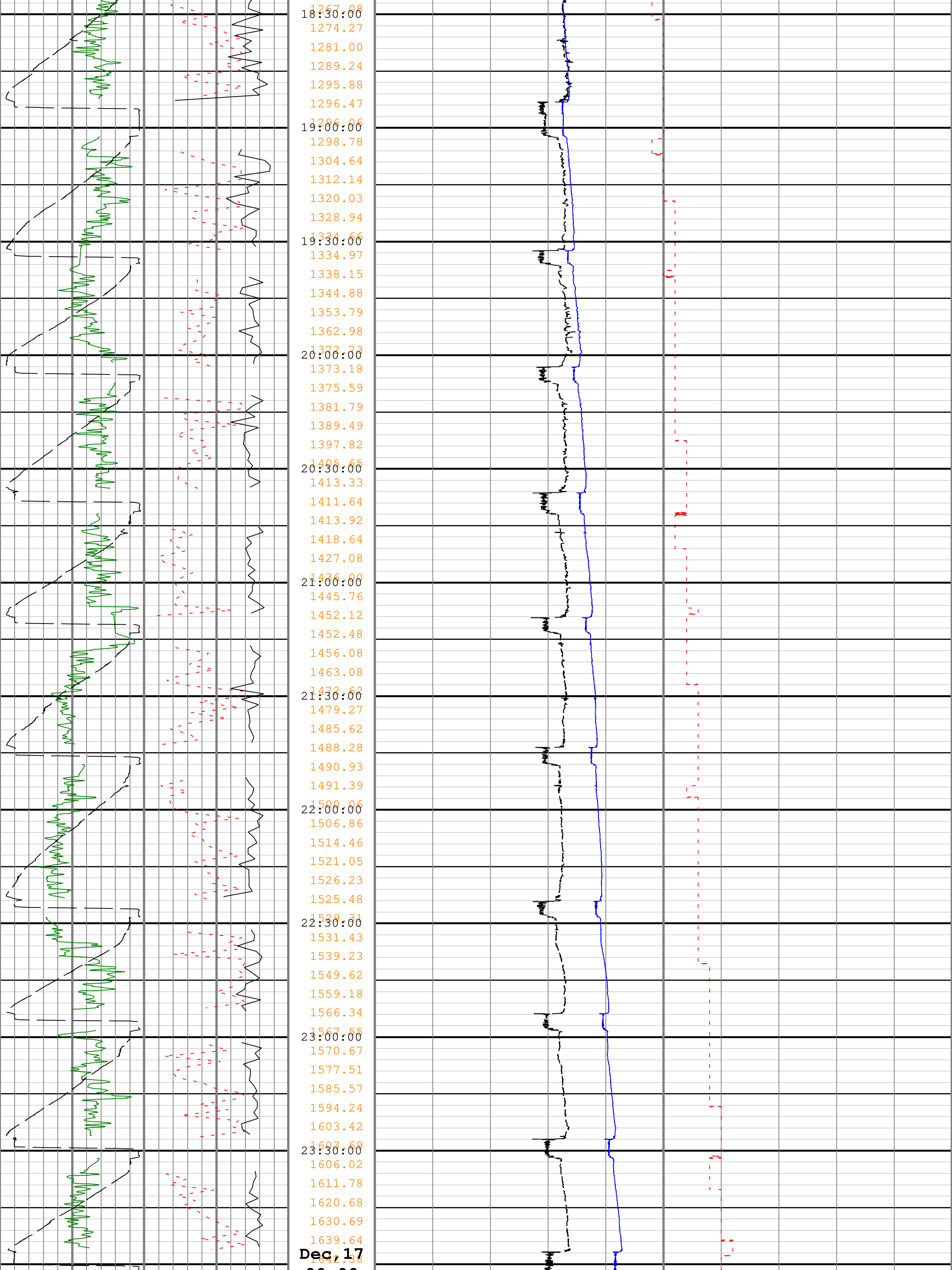
1257.79
1261.10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19:00:00
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1328.94
19:30:00
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1344.88
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1362.98
20:00:00
1373.18
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20:30:00
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1577.51
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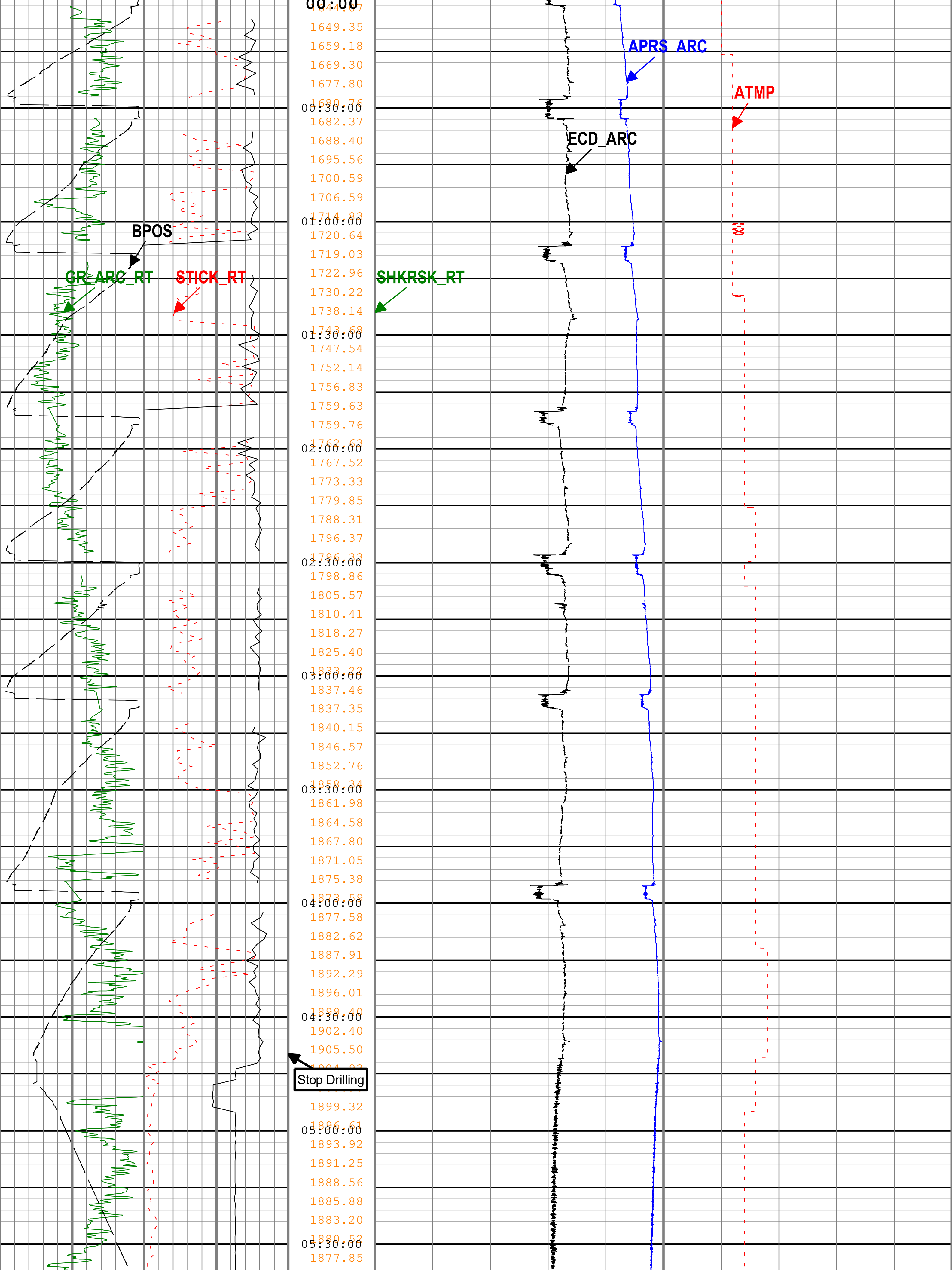
Dec, 17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1706.59
01:00:00
1720.64
1719.03
1722.96
1730.22
1738.14
01:30:00
1747.54
1752.14
1756.83
1759.63
1759.76
02:00:00
1767.52
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1892.29
1896.01
04:30:00
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1905.50
Stop Drilling
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05:00:00
1893.92
1891.25
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1885.88
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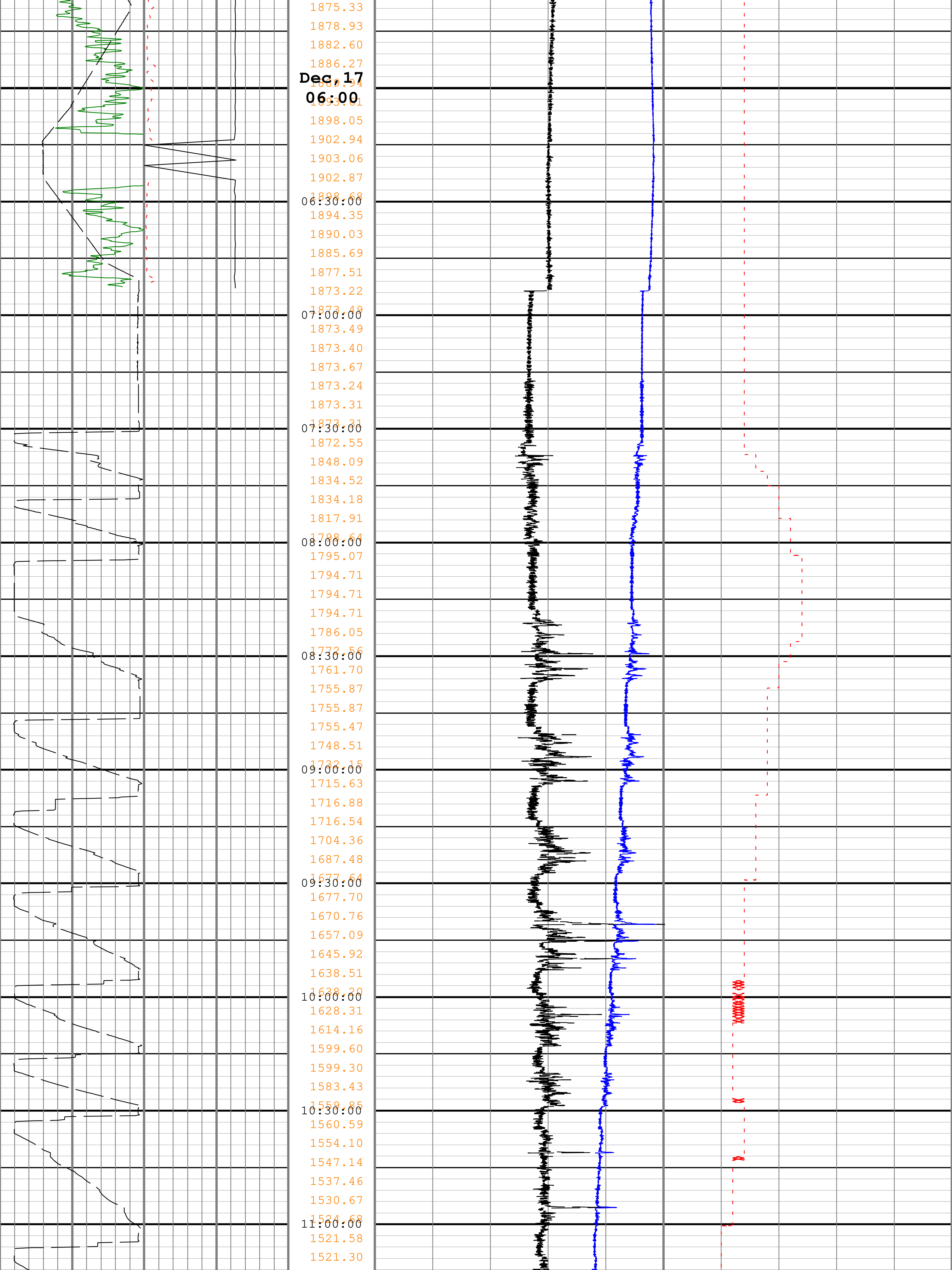
APRS_ARC
ATMP
ECD_ARC
BPOS
GR_ARC_RT
STICK_RT
SHKRSK_RT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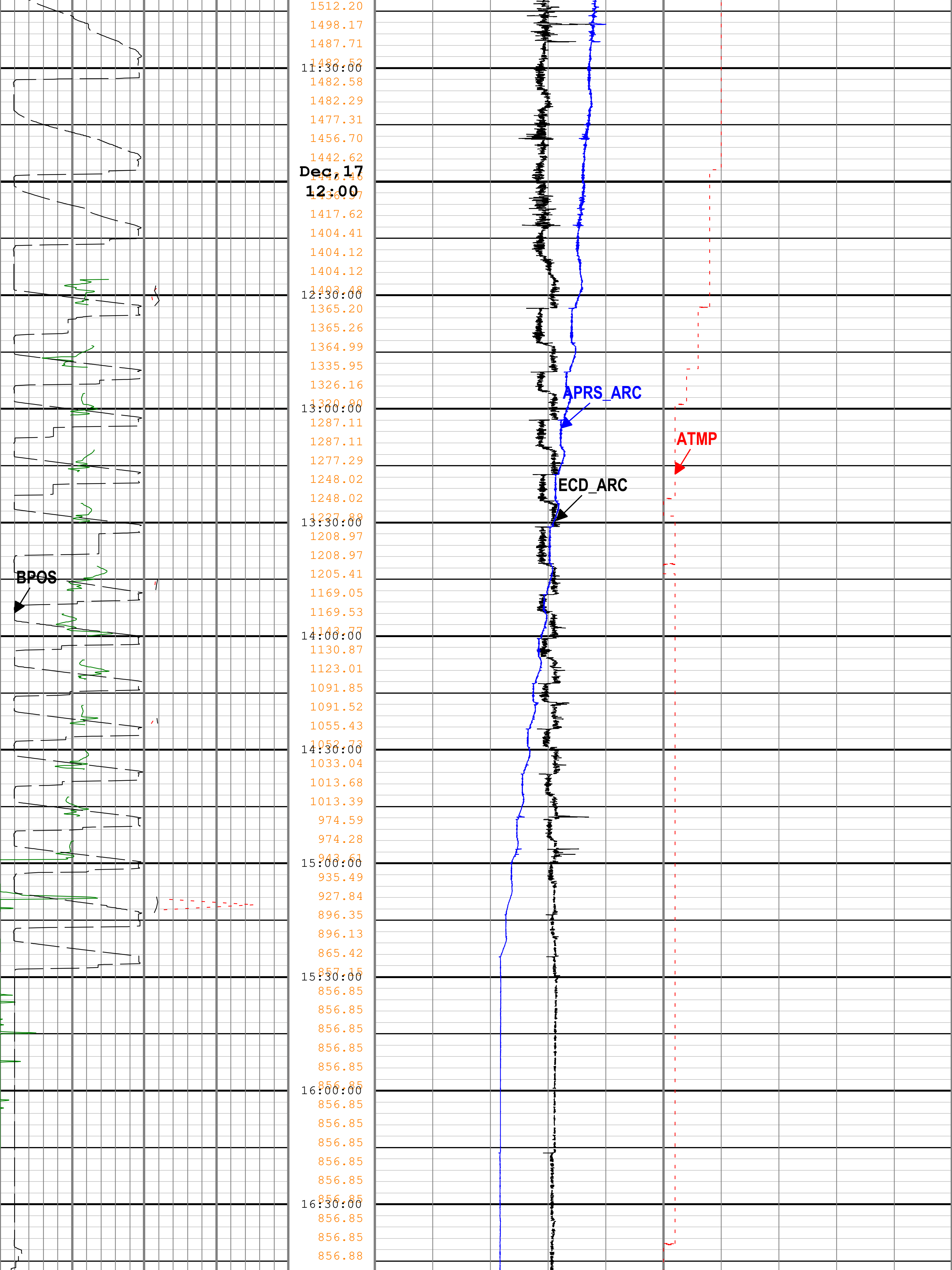
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11:30:00
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Dec,17
12:00
1417.62
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1364.99
1335.95
1326.16
APRS_ARC
13:00:00
1287.11
1287.11
ATMP
1277.29
1248.02
ECD_ARC
1248.02
13:30:00
1208.97
1208.97
1205.41
BPOS
1169.05
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14:00:00
1130.87
1123.01
1091.85
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14:30:00
1033.04
1013.68
1013.39
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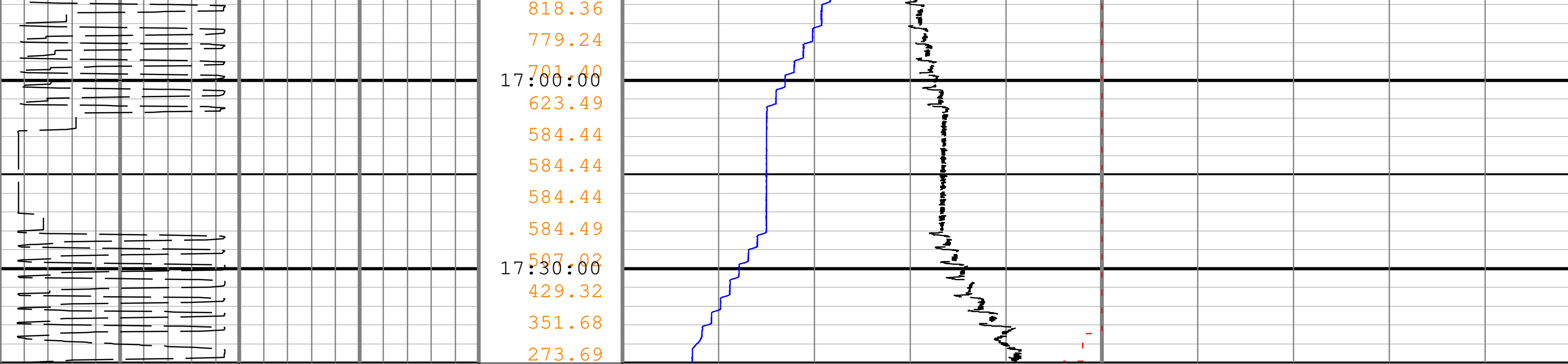

Gamma Ray (GR_ARC_RT) ARC9 RT
0 gAPI 150

Height of block above rig floor (BPOS) RT
0 m 45

Stick Slip Indicator (STICK_RT) TELE900 RT
0 c/min 500

Collar Rotational Speed (CRPM_RT) TELE900 RT
0 c/min 250

Depth Index (DEPTH) RT
m

Shock Risk (SHKRSK_RT) TELE900 RT
5 0

Shock Peak (SHKPK_RT) TELE900 RT
0 gn 250

Equivalent Circulating Density (ECD_ARC) ARC9 RM
1 g/cm3 2

Downhole Annulus Temperature (ATMP) ARC9 RM
0 degC 50

Downhole Annulus Pressure (APRS_ARC) ARC9 RM
0 bar 500

Description: Format: Log (Drilling Mechanics Light) Index Scale: 5 cm per 3600 s Index Type: Time Creation Date: 20-Dec-2019 14:32:04

Company:	Equinor
Well:	31/5-7
Field:	Eos
Rig Name:	West Hercules
Country:	Norway

Schlumberger

Drilling Mechanics - APWD

Time Based

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